ScheduleSync — 5-Minute Demo Video Script

Project: SIH26122 | Oil India Limited / Numaligarh Refinery Expansion (NRL)
Structure: 3:00 Min Admin / Planner Cockpit + 2:00 Min Field Supervisor PWA | 30-Second Exact Timestamps

Timeline	Panel / Mode	Focus Module	Primary Core Capability Demonstrated
0:00 - 0:30	Admin / Cockpit	Schedule Ingestion	Primavera P6 XER/XLSX ingestion, WBS hierarchy, 36 activities
0:30 - 1:00	Admin / Cockpit	AI Review Queue	3-Signal Hybrid matching (45% Semantic, 35% Entity, 20% WBS)
1:00 - 1:30	Admin / Cockpit	Match Deep-Dive	Side-by-side audit, field audio/OCR evidence, P6 baseline write-back
1:30 - 2:00	Admin / Cockpit	Gantt & Network	Interactive Frappe Gantt, L5/L6 node graph, sequence violation check
2:00 - 2:30	Admin / Cockpit	Analytics & EVM	Dynamic S-Curve (Planned vs Actual vs AI Forecast), SPI/CPI velocity
2:30 - 3:00	Admin / Cockpit	HSE, Tasks & RAG	AI Blocker tagging, supervisor rebalancing, historical lessons learned
3:00 - 3:30	Supervisor PWA	Field Mobile UX	Outdoor sunlight contrast, discipline isolation, 1-tap blocker report
3:30 - 4:00	Supervisor PWA	Voice & Translation	Regional Hindi speech transcription, auto-translation to English, audio file
4:00 - 4:30	Supervisor PWA	OCR & Guardrails	Physical DPR/photo OCR extraction, completed task anti-tamper lock
4:30 - 5:00	Supervisor PWA	Offline Resilience	IndexedDB offline queue, background sync on reconnect, closing wrap-up

PART 1: ADMIN & CENTRAL PLANNER COCKPIT (TIMELINE: 0:00 — 3:00)

0:00 - 0:30 — Schedule Baseline Ingestion & Cockpit Setup

Route: /admin/login -> /planner/ingestion

VISUAL ACTION ON SCREEN:
Log in with Planner credentials. Open Project Ingestion page. Select 'Numaligarh Refinery Expansion (Unit 3 & Offsites)'. Showcase the uploaded Primavera P6 schedule baseline parsing 36 activities, WBS codes, and predecessor dependencies.

EXACT SPOKEN SCRIPT (READ OUT LOUD):
"Hello everyone. Welcome to ScheduleSync, our intelligent schedule reconciliation platform built for mega-scale industrial capital projects like Oil India's Numaligarh Refinery Expansion. In massive infrastructure projects, central planners and site engineers often work in silos. Weekly site updates arrive late in scattered formats, causing the master Primavera P6 schedule to drift from reality. We solve this right here in the Planner Cockpit. We begin by ingesting the project baseline schedule directly from Primavera P6 files, automatically parsing thirty-six activities, WBS levels, and complex dependency relationships in seconds."

HACKATHON JUDGE KEY HIGHLIGHT:
Instant Primavera P6 XER/XLSX ingestion; creates structured SQLite/PostgreSQL pgvector schema.

0:30 - 1:00 — AI Review Queue & 3-Signal Hybrid Matcher	**Route: /planner/review**
VISUAL ACTION ON SCREEN:	*Navigate to Dashboard / Review Queue. Point mouse to top stat badges (Total Reports, In Review, Held, Verified). Hover over an incoming report with 86% Review confidence. Filter tabs by Piping and Civil disciplines.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Next, let's look at the heart of our intelligence: the Review Queue. As site updates stream in, our 3-Signal Hybrid Matching Engine evaluates each report against the master schedule. It calculates a unified confidence score using forty-five percent semantic vector similarity, thirty-five percent Gemini entity extraction, and twenty percent WBS metadata. High-confidence reports above ninety percent auto-verify immediately. Ambiguous reports between seventy and ninety percent are brought here into this review queue, allowing planners to verify critical work without drowning in hundreds of manual daily logs."
HACKATHON JUDGE KEY HIGHLIGHT:	3-Signal matching architecture (45/35/20 split) prevents false positives and ensures data integrity.

1:00 - 1:30 — Match Deep-Dive, Audit Trail & P6 Write-Back	**Route: /planner/review/:id**
VISUAL ACTION ON SCREEN:	*Click 'Review Match' on a Piping activity. Show side-by-side comparison: left card shows supervisor audio player, photo, and extracted quantities (45m, spool tags); right card shows Primavera P6 baseline activity. Click green 'Approve & Commit to P6' button.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Opening an item in the queue displays a full side-by-side audit view. On the left, we see the raw field evidence submitted by the supervisor: their original audio recording, site photos, and the entities our AI extracted, like pipe lengths, weld counts, and spool numbers. On the right, the platform identifies the exact matching baseline activity. When I click 'Approve', progress immediately writes back to the schedule, locking in an immutable audit trail and eliminating manual data entry errors forever."
HACKATHON JUDGE KEY HIGHLIGHT:	Direct write-back to baseline activities while storing complete, immutable multi-modal field evidence.

1:30 - 2:00 — Interactive Gantt & L5/L6 Dependency Network	**Route: /planner/schedule -> /planner/network**
VISUAL ACTION ON SCREEN:	*Click 'Gantt' in sidebar to show real-time Gantt timeline bars. Then click 'Dependency Network'. Pan across the interactive node graph. Hover over predecessor links and point to a highlighted red node showing a Sequence Violation alert.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Now, let’s see how this updates the overall project timeline. In our interactive Gantt view, planners can track real-time progress across Piping, Civil, and Electrical disciplines against original baselines. Even more powerful is our L5 and L6 Dependency Network. This interactive graph maps all predecessor and successor relationships. If a site team tries to mark an activity finished before its required predecessor is verified, our system instantly triggers a Sequence Violation warning, preventing out-of-order execution and costly site rework."
HACKATHON JUDGE KEY HIGHLIGHT:	Enforces critical path logic constraints (Finish-to-Start) and alerts planners to out-of-sequence reporting.

2:00 - 2:30 — S-Curve Analytics, EVM Metrics & AI Forecasting	**Route: /planner/analytics**
VISUAL ACTION ON SCREEN:	*Click 'Analytics' in sidebar. Point cursor to S-Curve chart (Planned curve, Actual Earned curve, AI Predictive Forecast line). Hover over EVM cards: SPI (Schedule Performance Index), CPI, and Delay Forecasting table.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Under Analytics, project leadership gets instant executive clarity. Our live S-Curve plots three critical curves: Planned Baseline, Actual Progress Earned, and our AI-powered Predictive Forecast line. Along with Earned Value Management metrics like Schedule Performance Index and Cost Performance Index, our forecasting model analyzes historical crew velocities. This flags potential delay risks and completion date overruns weeks in advance, giving project directors time to intervene proactively."
HACKATHON JUDGE KEY HIGHLIGHT:	Earned Value Management (EVM) combined with crew-velocity generative completion forecasting.

2:30 - 3:00 — HSE Blockers, Workload Balancing & RAG Memory	**Route: /planner/complaints -> /planner/workload -> /planner/historical**
VISUAL ACTION ON SCREEN:	*Click 'HSE & Blockers' to show AI auto-tagged tickets (Material shortage, Crane breakdown). Click 'Assignments' to view supervisor task distribution. Click 'History' and perform a quick search for past lessons learned.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Finally, our admin cockpit connects the operational dots. Site issues reported from the field are automatically categorized here into HSE, Material, or Equipment blockers with severity rankings. Planners can reassign tasks across supervisors using our Workload Balancing tool, and query our Historical Memory system using natural language to search past shutdown lessons and SOP documents. Now, let's see how easy it is for site engineers on the ground to capture this data using our mobile app."
HACKATHON JUDGE KEY HIGHLIGHT:	Comprehensive closed-loop governance: site tickets, workload rebalancing, and RAG knowledge retrieval.

PART 2: FIELD SUPERVISOR MOBILE PWA (TIMELINE: 3:00 — 5:00)

3:00 - 3:30 — Field PWA Setup & Assigned Workload	**Route: /supervisor/login -> /supervisor (Mobile View)**
VISUAL ACTION ON SCREEN:	*Switch browser to mobile responsive mode (iPhone 14 / 390x844px). Log in as 'Supervisor 1 - Piping'. Point out high-contrast sunlight UI and PWA install prompt. Scroll through assigned piping tasks with progress rings. Point to 'Report Blocker' button.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Now we switch directly into the boots of a site supervisor on the ground. This is our Field Progressive Web App. It is designed specifically for harsh, outdoor industrial environments, featuring high-contrast sunlight visibility, large touch targets, and full installability on any smartphone. When the supervisor logs in, they only see tasks specifically assigned to their discipline. They can instantly view remaining quantities, scheduled dates, and tap one button to raise urgent site blockers without navigating complex menus."
HACKATHON JUDGE KEY HIGHLIGHT:	Discipline and role isolation; high-contrast outdoor UI ergonomics; installable PWA.

3:30 - 4:00 — Voice Logging & Real-Time Hindi-to-English Translation	**Route: /supervisor/log**
VISUAL ACTION ON SCREEN:	*Tap '+' (Log Progress) bottom nav button. Select activity 'L6-PIP-101'. Switch language toggle to Hindi. Tap microphone to show Hindi dictation. Click 'Translate to English' to show instant translation while retaining original voice note audio file.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Logging progress on a busy construction site takes just seconds. Supervisors don’t need to type lengthy reports on small screens. They can simply tap the microphone and speak naturally in their regional language. Notice how the supervisor dictates in Hindi: 'Aaj humne Unit 3 mein 45 meter Line 24 pipeline ka kaam poora kar liya hai'. Our integrated AI instantly transcribes the speech, translates it into standardized technical English for Primavera P6, and attaches the original voice recording as verifiable site evidence."
HACKATHON JUDGE KEY HIGHLIGHT:	Real-time regional speech recognition (Faster-Whisper) + automatic LLM translation into EPC terminology.

4:00 - 4:30 — Document OCR & Completion Guardrails	**Route: /supervisor/log -> /supervisor/files**
VISUAL ACTION ON SCREEN:	*Attach a site photo or DPR scan. Demonstrate that selecting an already completed (100%) task shows the 'Activity Completed - Locked' safety warning badge. Select an active task, hit 'Submit', and display the immediate match confidence card.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Field supervisors can also snap photos of physical DPR report sheets, weld inspection tags, or equipment nameplates. Our backend OCR extracts line numbers, spools, and quantities automatically. To prevent human error or premature claims, ScheduleSync includes built-in safety guardrails: completed activities are locked automatically to prevent duplicate reporting. Once submitted, the supervisor receives instant feedback with their extracted data and matching confidence tier."
HACKATHON JUDGE KEY HIGHLIGHT:	OCR multimodal extraction + completion lock preventing fraudulent progress inflation.

4:30 - 5:00 — Offline Resilience & Grand Wrap-Up	**Route: /supervisor/log -> /supervisor/submissions**
VISUAL ACTION ON SCREEN:	*In Chrome DevTools, toggle network to 'Offline'. Show amber offline banner. Submit a test progress note; display 'Queued in IndexedDB'. Toggle network back to 'Online'; show instant automatic background sync toast. Open Submissions tab.*
EXACT SPOKEN SCRIPT (READ OUT LOUD):	"Refineries often have dead zones with zero cellular signal. ScheduleSync is engineered to never lose data. When network connectivity drops, the app switches to offline mode. All voice recordings and submissions are securely stored locally in an IndexedDB queue. As soon as the supervisor reconnects to Wi-Fi or cellular service, our background sync automatically transmits everything to the central server. ScheduleSync bridges ground truth with executive planning, turning scattered field notes into real-time schedule certainty. Thank you!"
HACKATHON JUDGE KEY HIGHLIGHT:	Zero data loss in offline field zones via client-side IndexedDB queue and Background Sync API.